

on the event study type of aggregation. Third, we make use of simultaneous inference procedures that explicitly account for potential multiple-testing problems; [Sun and Abraham \(2020\)](#) focuses on pointwise inference. On the other hand, we do not have any results highlighting the pitfalls associated with using TWFE specifications with leads and lags of treatment indicators to conduct causal inference; these are unique to [Sun and Abraham \(2020\)](#).

We also note that [Athey and Imbens \(2018\)](#) considers a staggered treatment adoption setup similar to ours. However, the starting point of [Athey and Imbens \(2018\)](#) is an assumption that the treatment adoption date is fully randomized which is stronger than our parallel trends assumptions. We also note that [Athey and Imbens \(2018\)](#) abstracts away from the important role played by covariates in the DiD analysis and does not consider aggregation schemes to summarize treatment effect heterogeneity like we do. On the other hand, we stress that the main focus of their paper is on providing design-based inference procedures for staggered DiD setups with random treatment dates. Their design-based inference procedures complement our sampling-based inference procedures.

Organization of the paper: The remainder of this article is organized as follows. Section 2 presents our main identification results. We discuss our different aggregation schemes in Section 3. Estimation and inference procedures for the treatment effects of interest are presented in Section 4. We revisit the effect of minimum wage on employment in Section 5. Section 6 concludes. Proofs as well as additional methodological results are reported in the Appendix. In the Supplementary Appendix, we present proofs for the results when only repeated cross-sections data is available, provide additional details about the empirical application, and present a small scale Monte Carlo simulation to illustrate the finite sample properties of our proposed estimators.²

2 Identification

2.1 Setup

We first introduce the notation we use throughout the article. We consider the case with $\mathcal{T}$ periods and denote a particular time period by t where $t = 1, \dots, \mathcal{T}$. In a canonical DiD setup, $\mathcal{T} = 2$ and no one is treated in period $t = 1$. Let $D_{i,t}$ be a binary variable equal to one if unit i is treated in period t and equal to zero otherwise. We make the following assumption about the treatment process:

Assumption 1 (Irreversibility of Treatment). $D_1 = 0$ *almost surely (a.s.)*. For $t = 2, \dots, \mathcal{T}$,

$$D_{t-1} = 1 \text{ implies that } D_t = 1 \text{ a.s..}$$

Assumption 1 states that no one is treated at time $t = 1$, and that once a unit becomes treated, that unit will remain treated in the next period.³ This assumption is also called staggered treatment

²Supplementary Appendix is available at https://pedrohcgs.github.io/files/Callaway_SantAnna_2020_supp.pdf.

³In applications, it can be the case that some units are already treated by the first time period. In our case, we would drop these units; this is analogous to the case with two time periods. The reason to drop these units is that untreated potential outcomes are never observed for this group which will imply that treatment effects are not identified for this group nor are they useful as a comparison group under a parallel trends assumption.

adoption in the literature. We interpret this assumption as if units do not “forget” about the treatment experience.⁴

Define G as the time period when a unit first becomes treated. Under Assumption 1, for all units that eventually participate in the treatment, G defines which “group” they belong to. If a unit does not participate in any time period, we arbitrarily set $G = \infty$. We define G_g to be a binary variable that is equal to one if a unit is first treated in period g (i.e., $G_{i,g} = \mathbf{1}\{G_i = g\}$) and define C to be a binary variable that is equal to one for units that do not participate in the treatment in any time period (i.e., $C_i = \mathbf{1}\{G_i = \infty\} = 1 - D_{i,\mathcal{T}}$). Let $\bar{g} = \max_{i=1,\dots,n} G_i$ be the maximum G in the dataset. Next, denote the generalized propensity score as $p_{g,s}(X) = P(G_g = 1 | X, G_g + (1 - D_s)(1 - G_g) = 1)$. Note that $p_{g,s}(X)$ indicates the probability of being first treated at time g , conditional on pre-treatment covariates X and on either being a member of group g (in this case, $G_g = 1$) or a member of the “not-yet-treated” group by time s (in this case, $(1 - D_s)(1 - G_g) = 1$). Many of our results use a specialized version of this generalized propensity score, and, henceforth, we define $p_g(X) = p_{g,\mathcal{T}}(X) = P(G_g = 1 | X, G_g + C = 1)$ which is the probability of being first treated in period g conditional on covariates and either being a member of group g or not participating in the treatment in any time period. Let $\mathcal{G} = \text{supp}(G) \setminus \{\bar{g}\} \subseteq \{2, 3, \dots, \mathcal{T}\}$ denote the support of G excluding $\bar{g}$.⁵ Likewise, let $\mathcal{X} = \text{supp}(X) \subseteq \mathbb{R}^k$ denote the support of the pre-treatment covariates. Finally, for a generic $\delta \geq 0$, let $\mathcal{G}_\delta = \mathcal{G} \cap \{2 + \delta, 3 + \delta, \dots, \mathcal{T}\}$.

Next, we set up the potential outcomes framework. Here, we combine the dynamic potential outcomes framework of Robins (1986, 1987) with the multi-stage treatment adoption setup discussed by Heckman et al. (2016); see also Sianesi (2004). Let $Y_{i,t}(0)$ denote unit i ’s untreated potential outcome at time t if they remain untreated through time period $\mathcal{T}$; i.e., if they were not to participate in the treatment across all available time periods. For $g = 2, \dots, \mathcal{T}$, let $Y_{i,t}(g)$ denote the potential outcome that unit i would experience at time t if they were to first become treated in time period g . Note that our potential outcomes notation accounts for potential dynamic treatment selection, though it also accommodates (pre-specified) treatment regimes (Murphy et al., 2001; Murphy, 2003). The observed and potential outcomes for each unit i are related through

$$Y_{i,t} = Y_{i,t}(0) + \sum_{g=2}^{\mathcal{T}} (Y_{i,t}(g) - Y_{i,t}(0)) \cdot G_{i,g} \quad (2.1)$$

In other words, we only observe one potential outcome path for each unit. For those that do not participate in the treatment in any time period, observed outcomes are untreated potential outcomes in all periods. For units that do participate in the treatment, observed outcomes are the unit-specific potential outcomes corresponding to the particular time period when that unit adopts the treatment.

We also impose the following random sampling assumption.

Assumption 2 (Random Sampling). $\{Y_{i,1}, Y_{i,2}, \dots, Y_{i,\mathcal{T}}, X_i, D_{i,1}, D_{i,2}, \dots, D_{i,\mathcal{T}}\}_{i=1}^n$ is independent and identically distributed (*iid*).

Assumption 2 implies that we have access to panel data; our results extend essentially immediately

⁴See Han (2020), de Chaisemartin and D’Haultfoeulle (2020) and Bojinov et al. (2020) for alternative setups where treatment can “turn off”.

⁵When there is a “never treated” set of units with $G = \infty$, $\mathcal{G}$ only excludes this group. When such “never-treated” group is not available, we exclude the latest-treated group as there will be no valid untreated comparison group for them.

to the case with repeated cross sections data and this case is developed in Appendix B. Here, we note that Assumption 2 allows us to view all potential outcomes as random. Furthermore, it does not impose restrictions between *potential outcomes* and treatment allocation, nor does it restrict the time series dependence of the observed random variables. On the other hand, Assumption 2 imposes that each unit i is randomly drawn from a large population of interest. For an alternative design-based inference approach, see [Athey and Imbens \(2018\)](#).

Henceforth, to keep the notation more concise, we will suppress the unit index i in our notation.

2.2 The Group-Time Average Treatment Effect Parameter

Given that different potential outcomes cannot be observed for the same unit at the same time, researchers often focus on identifying and estimating some average causal effects. For instance, in the canonical DiD setup with two time periods, the most popular treatment effect parameter of interest is the average treatment effect on the treated, denoted by⁶

$$ATT = \mathbb{E}[Y_2(2) - Y_2(0)|G_2 = 1].$$

In this paper, we consider a natural generalization of the ATT that is suitable to setups with multiple treatment groups and multiple time periods. More precisely, we use the average treatment effect for units who are members of a particular group g at a particular time period t , denoted by

$$ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0)|G_g = 1],$$

as the main building block of our framework. We call this causal parameter the *group-time average treatment effect*.

Note that the $ATT(g, t)$ does not impose any restriction on treatment effect heterogeneity across groups or across time. Thus, focusing on the family of $ATT(g, t)$'s allow us to analyze how average treatment effects vary across different dimensions in a unified manner. For instance, by fixing a group g and varying time t , one is able to highlight how average treatment effects evolve over time for that specific group. By doing this for different groups, we can have a better understanding about how treatment effect dynamics vary across groups. In addition, as we discuss in Section 3, one can build on the $ATT(g, t)$'s to form more aggregated causal parameters that are constructed to answer specific questions like: (a) What was the average effect of participating in the treatment across all groups that participated in the treatment by time period $\mathcal{T}$? (b) Are average treatment effects heterogeneous across groups? (c) How do average treatment effects vary by length of exposure to the treatment? (d) How do cumulative average treatment effects evolve over calendar time? We view this level of generality and flexibility as one of the main advantages of our framework that first focuses on the family of $ATT(g, t)$'s.

2.3 Identifying Assumptions

In order to identify the $ATT(g, t)$ and their functionals, we impose the following assumptions.

⁶Existence of expectations is assumed throughout.

Assumption 3 (Limited Treatment Anticipation). *There is a known $\delta \geq 0$ such that*

$$\mathbb{E}[Y_t(g)|X, G_g = 1] = \mathbb{E}[Y_t(0)|X, G_g = 1] \text{ a.s. for all } g \in \mathcal{G}, t \in \{1, \dots, \mathcal{T}\} \text{ such that } t < g - \delta.$$

Assumption 3 restricts anticipation of the treatment for all “eventually treated” groups. When $\delta = 0$, it imposes a “no-anticipation” assumption, see, e.g., [Abbring and van den Berg \(2003\)](#) and [Sianesi \(2004\)](#). This is likely to be the case when the treatment path is not a priori known and/or when units are not the ones who “choose” treatment status. However, Assumption 3 also allows for anticipation behavior, as long as we have a good understanding about the anticipation horizon δ . For instance, if units anticipate treatment by one period, Assumption 3 would hold with $\delta = 1$; see, e.g., [Laporte and Windmeijer \(2005\)](#) and [Malani and Reif \(2015\)](#) for the importance of accounting for potential anticipation behavior. Note that, under Assumption 3, $ATT(g, t) = 0$ for all pre-treatment periods such that $t < g - \delta$.

Next, we consider two alternative assumptions that impose restrictions on the evolution of untreated potential outcomes.

Assumption 4 (Conditional Parallel Trends based on a “Never-Treated” Group). *Let δ be as defined in Assumption 3. For each $g \in \mathcal{G}$ and $t \in \{2, \dots, \mathcal{T}\}$ such that $t \geq g - \delta$,*

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, G_g = 1] = \mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, C = 1] \text{ a.s..}$$

Assumption 5 (Conditional Parallel Trends based on “Not-Yet-Treated” Groups). *Let δ be as defined in Assumption 3. For each $g \in \mathcal{G}$ and each $(s, t) \in \{2, \dots, \mathcal{T}\} \times \{2, \dots, \mathcal{T}\}$ such that $t \geq g - \delta$ and $t + \delta \leq s < \bar{g}$,*

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, G_g = 1] = \mathbb{E}[Y_t(0) - Y_{t-1}(0)|X, D_s = 0, G_g = 0] \text{ a.s..}$$

Assumptions 4 and 5 are two different conditional parallel trends assumptions that generalize the two-period parallel trends assumption to the case where there are multiple time periods and multiple treatment groups; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#) and [Sant’Anna and Zhao \(2020\)](#). Both assumptions hold after conditioning on covariates X . This can be important in many applications in economics particularly in cases where there are covariate specific trends in outcomes over time and when the distribution of covariates is different across groups. For example, [Heckman et al. \(1997\)](#) motivates conditional parallel trends assumptions in the context of evaluating a job training program. For evaluating job training programs, the distribution of observed covariates such as age, employment history, and years of education is often quite different between individuals who participate in job training and those that do not. When the path of labor market outcomes (in the absence of participating in job training) depends on these covariates, a conditional parallel trends becomes more plausible than an unconditional parallel trends assumption. In fact, ignoring the presence of covariate-specific trends can result in important biases when evaluating causal effects of policy interventions using unconditional DiD methods.

Assumptions 4 and 5 differ from each other depending on the comparison group one is willing to use in a given application. More specifically, Assumption 4 states that, conditional on covariates, the average outcomes for the group first treated in period g and for the “never-treated” group would have followed parallel paths in the absence of treatment. Assumption 5 imposes conditional parallel trends between